



ResiliLytics Dashboard

Sourcing Intelligence for Resilient Supply Chains

ResiliLytics User Manual: SME-Native Supply Chain Resilience Intelligence Framework

Minimal-Input, Multidimensional Risk-to-Action Architecture

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ResiliLytics User Manual: SME-Native Supply Chain Resilience Intelligence Framework

1. Introduction — Why Resilience Must Be Structured

Supply chain disruption is no longer an exceptional event.

For Small and Medium Enterprises (SMEs), disruption is not merely operational inconvenience, it is structural vulnerability. Where large enterprises may absorb volatility through scale and diversification, SMEs often operate within tighter supplier networks, narrower geographic footprints, and thinner financial margins.

ResiliLytics was designed in response to that reality.

It was built on a simple but powerful premise:

Resilience should not require enterprise-level infrastructure.

From minimal supplier data, meaningful structural intelligence can be derived.

ResiliLytics transforms basic sourcing inputs into a structured resilience posture assessment and, critically, into sequenced mitigation guidance.

This is not a visualization tool alone.

It is a decision architecture.

2. Origin & Research Context

ResiliLytics was developed as an original research contribution focused specifically on SME supply chain resilience.

Existing resilience analytics systems frequently depend on:

- ERP integration
- Complex multi-system data ingestion
- Advanced enterprise analytics infrastructure

SMEs often lack these capabilities.

ResiliLytics demonstrates an alternative model:

Minimal Input

- Multidimensional Structural Inference
- SME-Calibrated Weighted Architecture
- Constraint Classification
- Risk-to-Action Sequencing
- Controlled Analytical Expansion

The framework was informed by documented SME disruption patterns, including:

- Supplier concentration fragility
- Geographic clustering exposure
- Volatility amplification effects
- Limited sourcing optionality
- Margin transmission sensitivity

Its contribution lies not in any single metric, but in the integrated architecture that connects structural exposure to sequenced action.

3. The Four-Layer Architecture

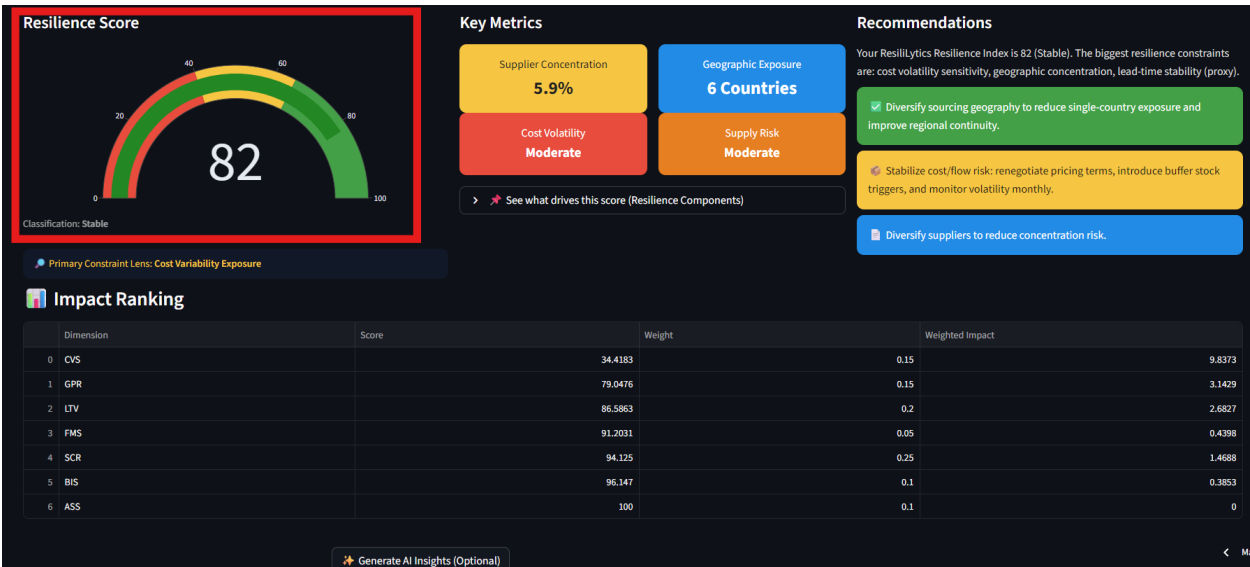
ResiliLytics operates across four integrated layers.

Layer 1 — Monitoring

The system calculates:

- A Composite Resilience Index (0–100)
- Seven structural resilience dimensions
- A weighted impact prioritization table

This provides immediate structural visibility.



Conceptual ResiliLytics Risk-to-Action Architecture

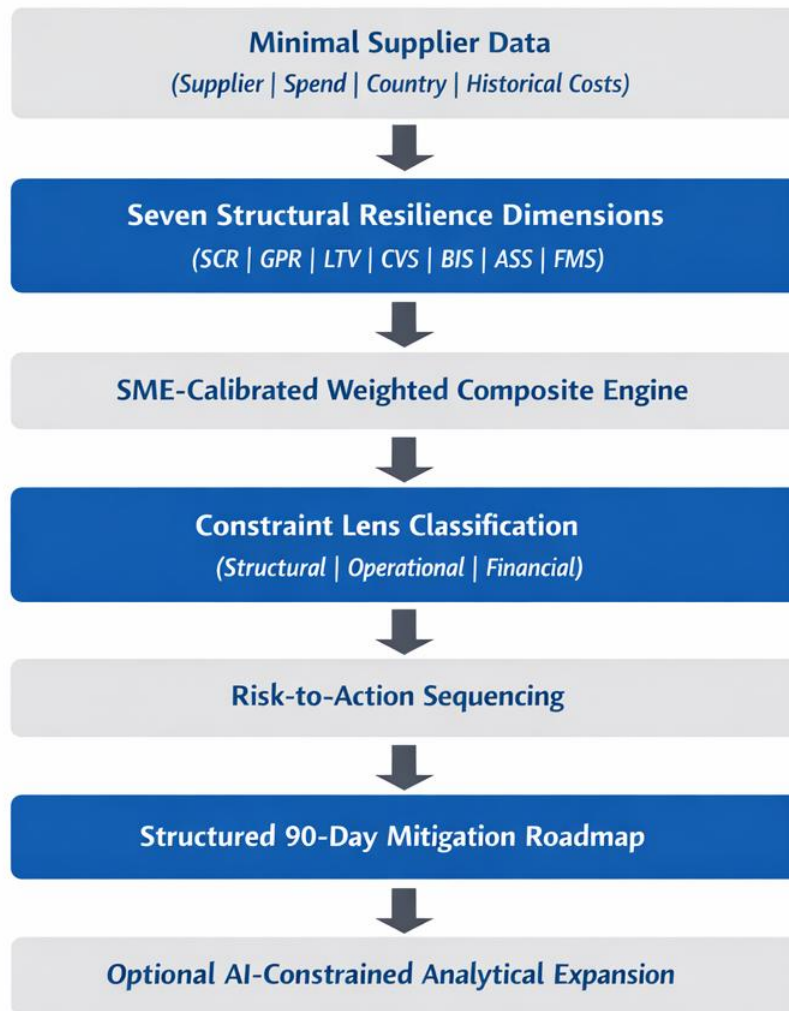


Figure 7. Conceptual ResiliLytics Risk-to-Action Architecture

Layer 2 — Diagnosis

Beyond visibility, the framework identifies:

- The weakest resilience dimensions
- The dominant systemic constraint lens
- The likely risk transmission pathway

This transforms observation into structural understanding.

AI Risk Interpretation & Action Plan

Executive Insight

Executive Summary: Composite Score 82 (Stable). Generally stable structural posture. Primary constraint lens: ****Financial/volatility exposure****. Key limiting dimensions: Cost Volatility Sensitivity, Geographic Portfolio Resilience, Lead-Time Stability Proxy.

Diagnosis: • **CVS — Cost Volatility Sensitivity**: Cost variability increases budgeting uncertainty and can trigger margin compression under shock conditions. • **GPR — Geographic Portfolio Resilience**: Single-country exposure increases disruption sensitivity (logistics, policy, FX, and regional shocks). • **LTV — Lead-Time Stability Proxy**: Timing variability (proxy) reduces planning reliability and increases reactive expediting and schedule churn.

>  Advanced Diagnostic Layer

Priority Mitigation Actions

Priority 1

Dampen cost volatility: monitor variance monthly, renegotiate pricing mechanics, and prioritize high-variance inputs for stabilization.

▼  Advanced Diagnostic Layer

Systemic Risk Classification

Structural Resilience Score: 91.1
Operational Stability Score: 91.4
Financial Exposure Score: 62.8

Primary Systemic Constraint: Financial Exposure

Risk Transmission Pathway

Identified Weakness: CVS
Likely Consequence: Margin volatility under cost fluctuation scenarios.
Intervention Direction: Stabilize pricing mechanisms and renegotiation bands.

Layer 3 — Prescriptive Automation

ResiliLytics automatically generates:

- Three prioritized mitigation actions
- A structured 90-day roadmap
- Sequenced intervention logic aligned with the dominant constraint

This is where the system moves from assessment to structured action.

 **Priority Mitigation Actions**

Priority 1

Dampen cost volatility: monitor variance monthly, renegotiate pricing mechanics, and prioritize high-variance inputs for stabilization.

Priority 2

Reduce single-country exposure: introduce at least one alternate sourcing geography for the most critical categories.

Priority 3

Improve operational stability: introduce disruption triggers, reliability scoring, and escalation rules for late/variable suppliers.

 **Structured 90-Day Resilience Roadmap**

0-30 Days: Tag high-variance inputs; begin monthly volatility tracking and isolate the top variance drivers.

30-60 Days: Renegotiate pricing mechanics (indexed clauses / volatility bands) for high-variance categories; define thresholds for action.

60-90 Days: Establish margin protection playbooks (substitutions, contract clauses, escalation) and review volatility exposure quarterly.

Layer 4 — Optional AI-Assisted Expansion

For deeper strategic interpretation, users may copy a structured analytical prompt into ChatGPT.

Important safeguards:

- No uploaded data is transmitted externally
- Only structured summary metrics are manually copied
- AI output does not override the model

 **Impact Ranking**

	Dimension	Score	Weight	Weighted Impact
0	CVS	34.4183	0.15	9.8373
1	GPR	79.0476	0.15	3.1429
2	LTV	86.5863	0.2	2.6827
3	FMS	91.2031	0.05	0.4398
4	SCR	94.125	0.25	1.4688
5	BIS	96.147	0.1	0.3853
6	ASS	100	0.1	0

 Generate AI Insights (Optional)

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Optional: Deep Analysis via ChatGPT

Copy this prompt into ChatGPT:

You are reviewing structured output from the ResiliLytics SME-Native Resilience Analytics Dashboard.

ResiliLytics is a minimal-input, weighted, multidimensional resilience modeling framework designed for Small and Medium Enterprises (SMEs).

The system calculates:

- A composite resilience index (0–100)
- Seven weighted resilience dimensions
- A structured 90-day mitigation pathway

[Open ChatGPT in New Tab](#)

4. Data Requirements

ResiliLytics intentionally minimizes data burden.

Required columns:

Column	Description
Supplier	Supplier name
Spend	Total spend allocation
Country	Supplier location
Historical_Costs	Semi-colon separated historical cost values

Example format:

12.5; 12.8; 13.1; 12.9

Historical cost data strengthens volatility-sensitive dimensions but is not mandatory.
All processing occurs within the active browser session. No uploaded data is stored.

5. The Seven Resilience Dimensions

Each dimension is normalized on a 0–100 scale.
Higher values represent stronger structural resilience.

1. Supplier Concentration Resilience (SCR)

Measures dependency on dominant suppliers.

2. Geographic Portfolio Resilience (GPR)

Captures exposure to single-country sourcing concentration.

3. Lead-Time Stability Proxy (LTV)

Derived structural indicator reflecting timing instability risk.

4. Cost Volatility Sensitivity (CVS)

Measures exposure to historical price variability.

5. Buffer Inventory Strength (BIS)

Proxy-based indicator of shock absorption capacity under volatility conditions.

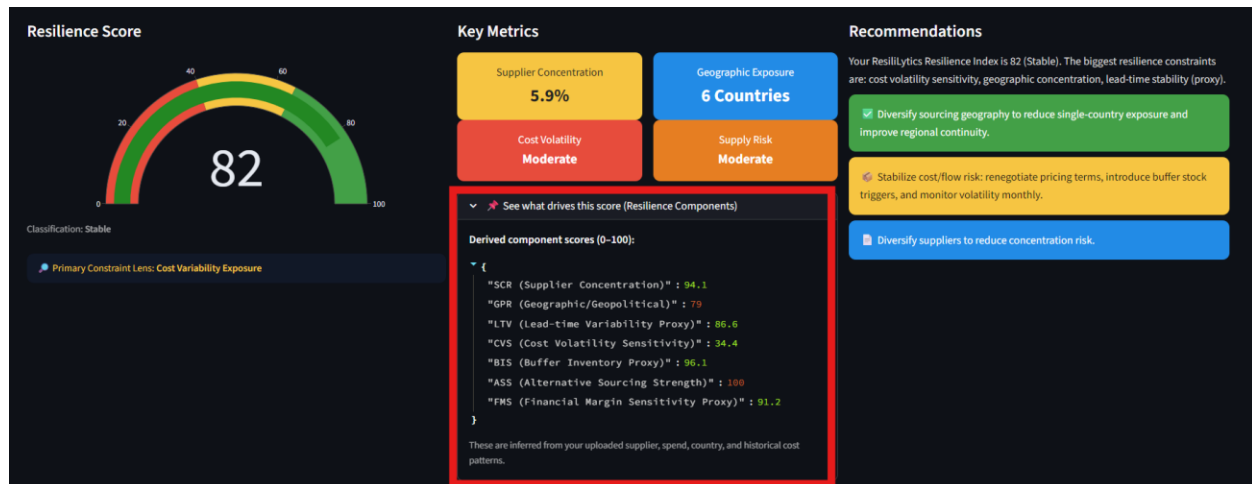
6. Alternative Sourcing Strength (ASS)

Reflects supplier optionality and switching flexibility.

7. Financial Margin Sensitivity (FMS)

Structural indicator of how volatility and concentration may transmit into profitability pressure.

These indicators are derived structural measures inferred from concentration and variability relationships within the dataset.



6. Weighted Composite Framework

The Composite Resilience Index integrates all seven dimensions using SME-calibrated weights:

SCR — 25%
GPR — 15%
LTV — 20%
CVS — 15%
BIS — 10%
ASS — 10%
FMS — 5%

These weights reflect empirically observed SME disruption dynamics in which concentration and timing instability often precede financial strain.

7. Adaptive Concentration Mode

When supplier concentration exceeds a structural fragility threshold (50%), the system activates adaptive weighting.

In this mode:

- Concentration emphasis increases
- Financial sensitivity emphasis increases
- Remaining weights adjust proportionally

This prevents linear averaging from understating high-dependency exposure.

Specific recalibration mechanics remain internal to the framework.

8. Understanding Your Score

Score Range	Classification
0–40	Critical Risk
41–65	Vulnerable
66–85	Stable
86–100	Resilient

The score reflects structural posture — not guaranteed future performance.

9. Impact Ranking

The Impact Ranking table displays:

- Raw dimension score
- Active weight
- Weighted impact contribution

This allows mitigation to be prioritized logically rather than reactively.



10. Risk-to-Action Sequencing Architecture

ResiliLytics is designed around a structured Risk-to-Action principle.

Identified weaknesses trigger:

1. Constraint classification
2. Consequence mapping
3. Prioritized mitigation sequencing
4. Structured 90-day action planning

This ensures that risk visibility translates into decision-ready intervention.

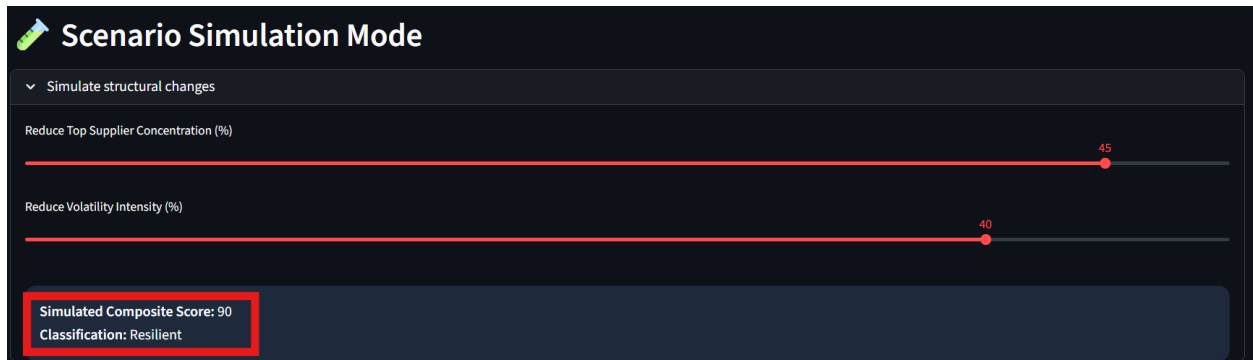
11. Scenario Simulation Mode

Users may simulate:

- Reduced supplier concentration
- Reduced volatility intensity

The system recalculates structural posture instantly.

This enables proactive strategic testing before implementation.



12. Model Assumptions & Boundaries

ResiliLytics:

- Measures structural exposure
- Does not forecast revenue
- Does not predict insolvency
- Does not guarantee disruption immunity

Certain dimensions are proxy-based structural indicators derived from interaction relationships within the dataset.

The framework is calibrated specifically for SME environments.

13. Data Privacy & Security

- No uploaded data is stored
- No session data is retained
- No automatic external transmission occurs
- AI expansion requires manual metric copying

ResiliLytics operates as a non-commercial research prototype.

14. Intended Use & Contribution

ResiliLytics demonstrates that meaningful resilience intelligence can be derived from minimal structured input.

Its contribution lies in system architecture:

Minimal Input

- Multidimensional Structural Inference
- Adaptive Weighting
- Constraint Classification
- Risk-to-Action Sequencing
- Controlled Generative Expansion

It is intended for strategic evaluation, research demonstration, and resilience planning within SME environments.